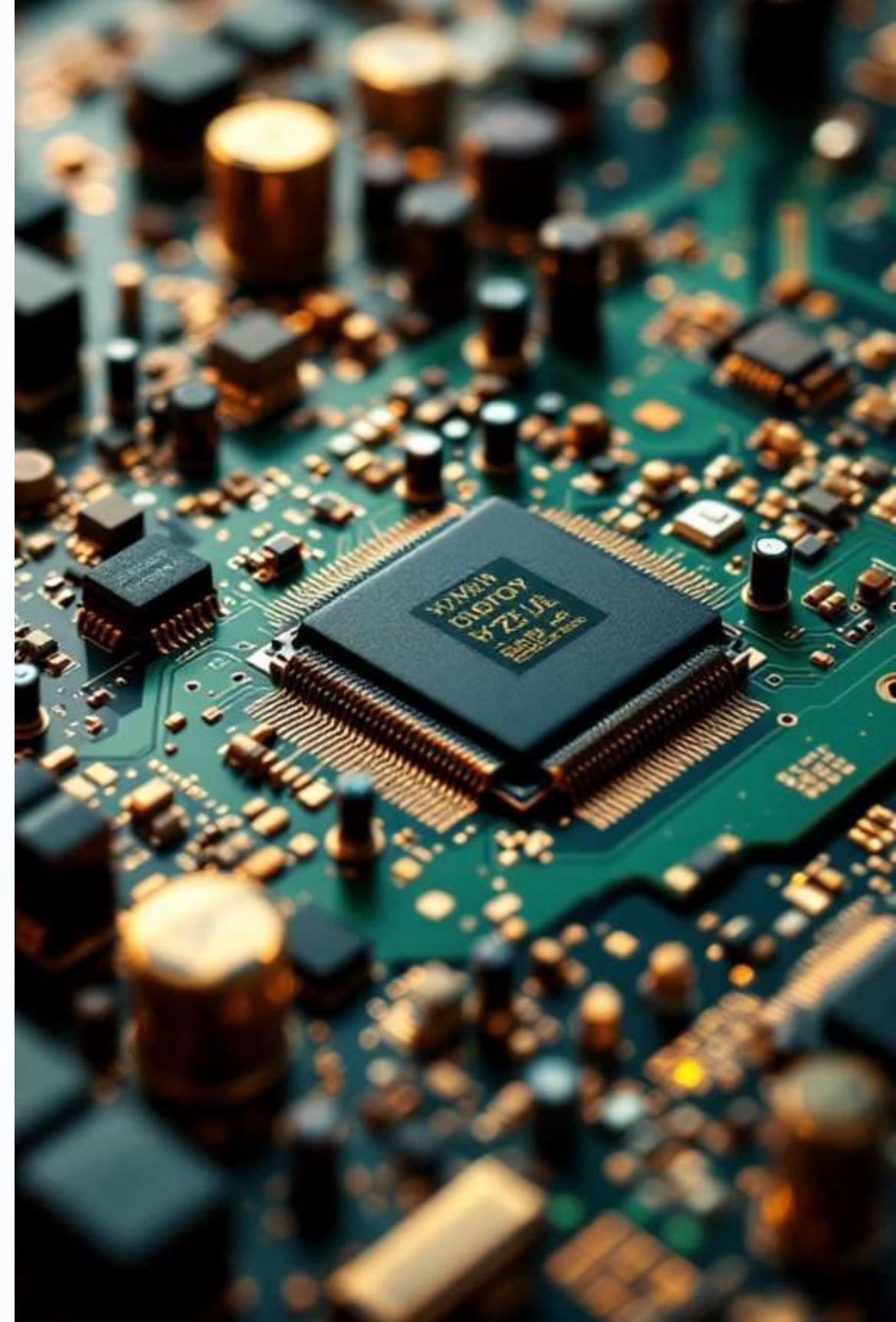


CircuitGuard — PCB Defect Detection & Classification

Automated PCB Inspection using Template-Subtraction, Contouring &
EfficientNetB4, Frontend and Backend Integration.

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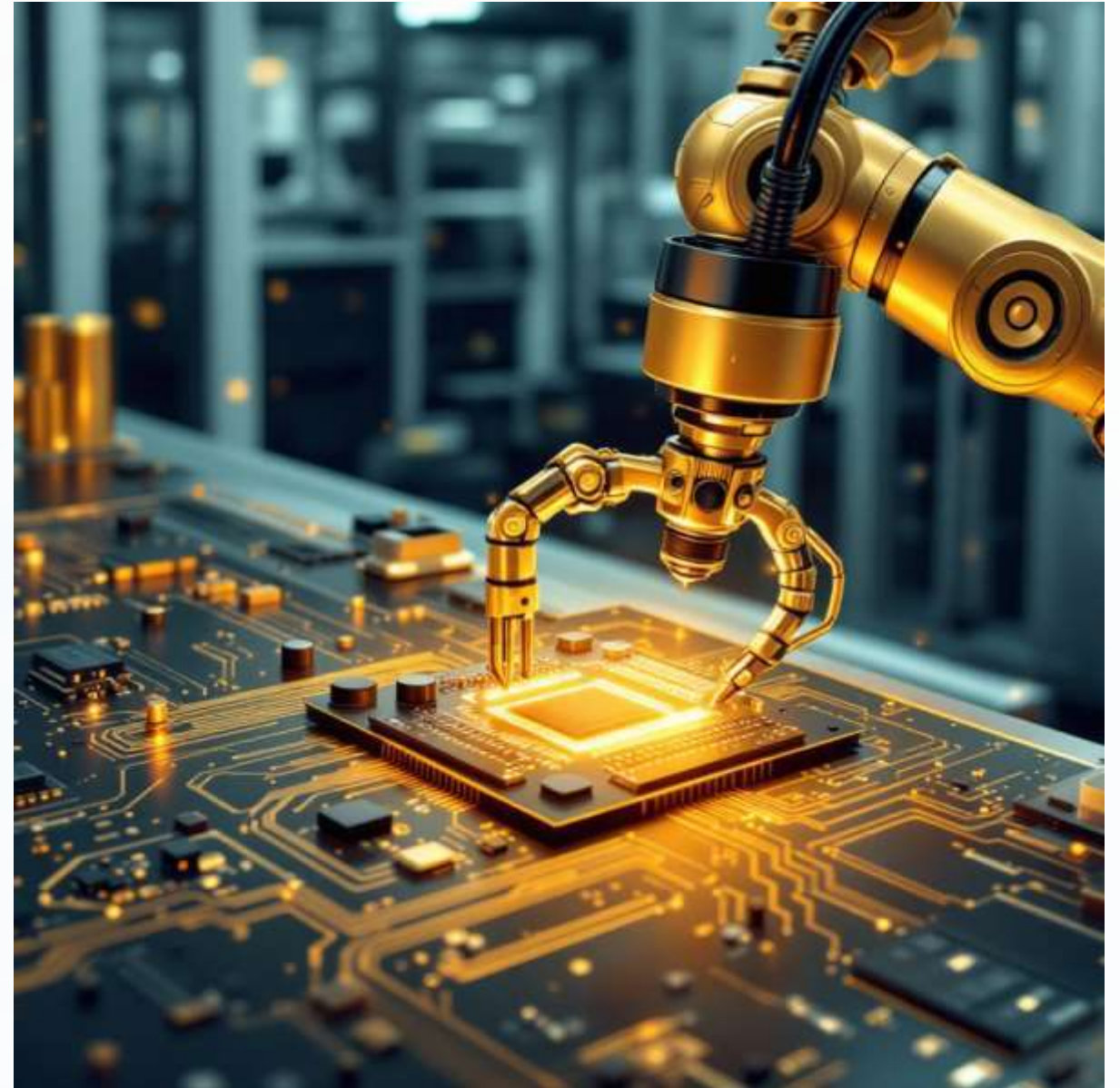


Introduction: The Need for Precision in PCB Quality

Printed Circuit Boards (PCBs) are the foundation of all modern electronic devices. A single, small defect can lead to expensive product rework, catastrophic field failures, and significant reputation damage.

- ❏ Traditional manual inspection methods are **slow, prone to human error, and highly subjective**. The goal of CircuitGuard is to introduce scalable, automated quality assurance.

This solution combines the robustness of classical Computer Vision (CV) with the classification power of a state-of-the-art Convolutional Neural Network (CNN) to achieve both localization and precise identification of defects.



Addressing the Core Problem Statement

Current quality assurance often fails due to limitations in detection capability and subjectivity. We must solve two critical pain points in one integrated system.



Subtle Defect Identification

Manual inspection and rudimentary Automated Optical Inspection (AOI) systems frequently miss complex or subtle defects, such as fine scratches or minor pinholes, necessitating time-consuming human verification.



Diversity of Defect Types

The system must accurately handle a wide range of common PCB faults, including missing hole, short, open circuit, spurious copper, mouse bite and spur.



Traceability & Latency

We need high-accuracy localization and classification, delivered with low latency, and results must be fully exportable for quality control logs and regulatory compliance.

Project Objectives: Defining the CircuitGuard Goal

The primary objective is to build a robust, production-ready pipeline that delivers high accuracy and speed for automated PCB inspection.



Defect Localization

Accurately detect and pinpoint defects using classical template subtraction and contour extraction techniques.



Precise Classification

Classify each isolated defect into predefined categories utilizing a pre-trained EfficientNet-B4 model for robust feature extraction.



Web Visualization & Export

Deliver a functional web user interface (UI) enabling easy image upload, defect visualization, and export of annotated images and comprehensive CSV log files.



Performance Benchmarks

Achieve near real-time processing (target: ≤ 5 seconds per image) while maintaining a high classification accuracy of **94%**.

Methodology

CircuitGuard employs a sequential five-step process, leveraging both image processing and deep learning for maximum performance.



Subtraction & Masking

Perform image subtraction to create a difference map. Apply Otsu thresholding and morphological operations to create a clean, binary mask highlighting only defect regions.



Contour Detection

The process of identifying and extracting the outlines and boundaries of objects on a circuit board.



ROI Extraction

Detect contours in the mask, filter by size, and crop standardized 128×128 Regions of Interest (ROIs) for classification.



EfficientNet Training

Train a pre-configured EfficientNet-B4 model via transfer learning (PyTorch, timm). Implement data augmentation and class balancing for improved generalization.

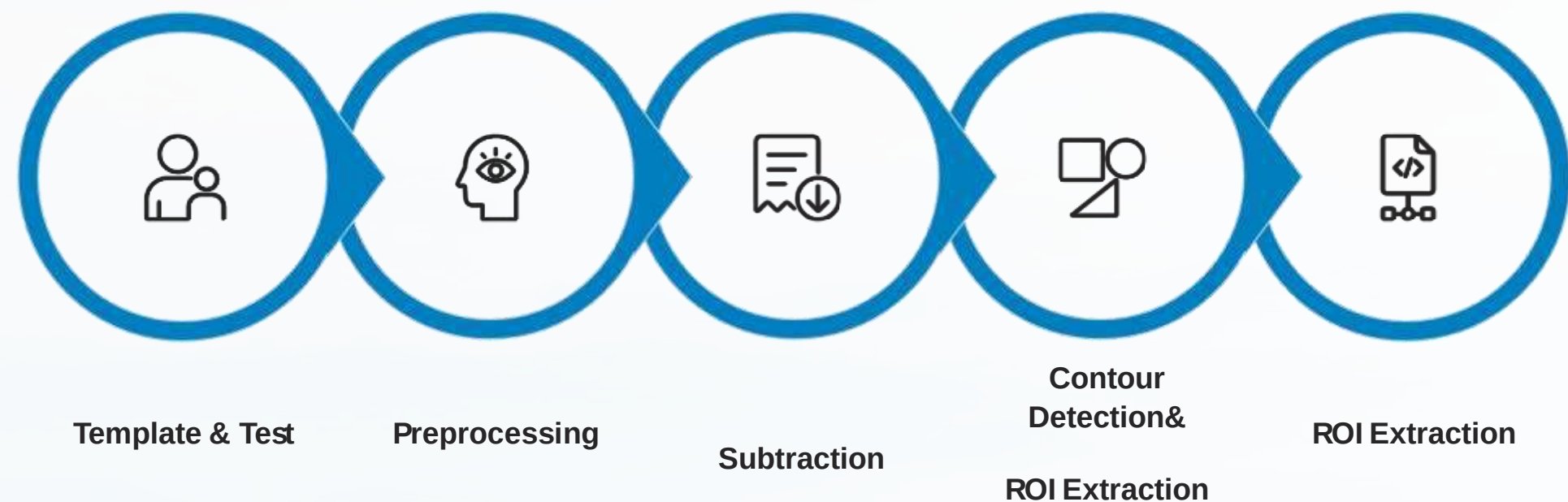


Inference & Annotation

Run ROIs through the trained model, map the predicted defect class back to the original full image, and output the annotated image with bounding boxes and labels.

System Architecture Overview

The system is designed as a modular, three-tiered architecture focused on efficient data flow and clear separation of concerns.



Backend Services

Modular Python services manage image preprocessing (OpenCV), model inference (PyTorch), and result generation/storage.

Classification Engine

The core deep learning service containing the EfficientNet model checkpoint, handling high-throughput classification of ROIs.

Frontend Interface

A user-friendly Streamlit application providing upload capabilities, visualization of the annotated output, and controls for data export.

Core Pipeline Modules

Each stage of the CircuitGuard pipeline serves a specific, critical function, ensuring accurate defect isolation before classification.

- **Subtraction & Masking**
Generates the difference image between test and template, applying thresholding and morphological operations to isolate defect areas.
- **Image Preprocessing**
Handles image alignment, photometric and geometric normalization, and initial resizing to prepare for subtraction.
- **ROI Pipeline**
Detects defect contours, filters based on size/ratio heuristics, calculates bounding boxes, and extracts standardized patches for the model.
- **Model Module**
Manages the EfficientNet model (training scripts, checkpoint loading, and high-speed inference wrapper).
- **UI Interface**
Develop Web UI for Image Upload and Visualization
- **Evaluation & Export**
Computes final performance metrics, compiles CSV logs of defect locations/classes, and overlays bounding boxes onto the original image for output.

Key Technologies and Toolkit

The CircuitGuard system relies on a stack of industry-standard, high-performance libraries across computer vision and deep learning.



OpenCV & NumPy

Used for all classical computer vision tasks: image alignment, template subtraction, thresholding, contour detection, and mask generation.



PyTorch & timm

The foundation for deep learning. PyTorch powers model training and inference, utilizing the `timm` library for accessing the EfficientNet-B4 architecture.



pandas & Matplotlib

Essential for data handling, metric computation, generating traceable CSV logs, and creating visual plots for performance analysis.



Streamlit

Used for rapid prototyping of the web user interface (UI), facilitating quick testing, visualization, and demonstration of the pipeline's end-to-end functionality.

Performance and Quality Analysis

Initial benchmarks demonstrate strong performance, with opportunities for further optimization, particularly in latency and metric validation.

94%

Classification Accuracy

Reported accuracy on the held-out test set, demonstrating the generalization capability of the EfficientNet-B4 model.

<5s

Inference Latency Target

Maximum acceptable processing time per full PCB image. Optimization via hardware acceleration or TorchScript is a potential next step.

For a deeper dive into model efficiency, a full confusion matrix and per-class precision/recall figures are computed and available in the technical appendix.

Conclusion

CircuitGuard successfully demonstrates a robust, hybrid approach to automated PCB quality assurance.



Integrated Solution

A functional pipeline combining classical CV for precise localization and EfficientNet for high-accuracy classification.



Enhanced Traceability

Provides exportable logs (CSV) and annotated images, crucial for audit trails and manufacturing process control.



Scalability

Designed with a modular, integration-friendly architecture suitable for deployment in high-throughput manufacturing environments.

This innovative system represents a significant leap forward in ensuring the integrity and reliability of printed circuit boards, moving beyond traditional manual inspection methods that are often prone to human error and inefficiency.

The successful development and validation of CircuitGuard highlight its capability to accurately detect a wide array of PCB defects, from subtle manufacturing flaws to critical functional issues. This achievement directly translates into tangible benefits for manufacturers, including reduced rework costs, improved product yield, and ultimately, higher overall product quality delivered to the market.

Demo

